

Introduction to uv

Outline

- What is uv?
- Installation
- Basic usage

What is uv?

An extremely fast Python package and project manager, written in Rust.

Installing Trio's dependencies with a warm cache.

Astral

- Astral 是一間專注於 Python tooling 的公司
- 主要使用 Rust
- 旗下的產品有: uv, ruff, ty, pyx(beta), ...
- rye 已經不再維護了

回憶

- 先接觸 ruff
- 後來觀望 uv
- uv 支援 project management 的功能後就開始使用了



考古

- [2024-02-15](#): uv 為 pip, pipx, venv 的替代品
- [2024-08-20](#): uv 加入專案管理的功能 (Cargo for Python), 可以取代 Poetry, PDM, Rye 等等
- [2025-12-16](#): ty beta release, 專注於 type checking 的工具, 取代 mypy, pyright 等等
- [2026-03-19](#): Astral 加入 OpenAI 的 Codex 團隊

ref: <https://astral.sh/blog>

印象中以前的習慣

- pip/venv + `requirements.txt` or `setup.py`
- conda + `environment.yml`
- poetry + `pyproject.toml`
- rye + `pyproject.toml`

當時為什麼改用 uv?

- 速度非常快
- 可以一次取代 pip, pipx, venv, pyenv, poetry 等等工具
- 有 cross-platform lockfile (uv.lock) 可重現度高
- 可以管理 python 版本
- 對 monorepo 支援很好 (workspace)

Installation

```
# macOS and Linux
curl -LsSf https://astral.sh/uv/install.sh | sh

# Windows (PowerShell)
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"

# Windows (Scoop)
scoop install main/uv
```

ref: <https://docs.astral.sh/uv/getting-started/installation>

Upgrading uv

不會太常用到，但也不是沒機會: [uv security advisory: ZIP payload obfuscation](#)

```
uv self update
```

```
# If you installed uv with pip, you can also upgrade it with pip:  
pip install --upgrade uv
```

ref: <https://docs.astral.sh/uv/getting-started/installation/#upgrading-uv>

Shell autocomplete

```
# zsh
echo 'eval "$(uv generate-shell-completion zsh)"' >> ~/.zshrc

# fish
echo 'uv generate-shell-completion fish | source' > ~/.config/fish/completions/uv.fish
```

ref: <https://docs.astral.sh/uv/getting-started/installation/#shell-autocompletion>

Uninstallation

```
# Clean up stored data (optional):  
uv cache clean  
rm -r "$(uv python dir)"  
rm -r "$(uv tool dir)"  
  
# macOS and Linux  
rm ~/.local/bin/uv ~/.local/bin/uvx
```

ref: <https://docs.astral.sh/uv/getting-started/installation/#uninstallation>

Running script

```
uv run example.py

# create a Python script (optional)
uv init --script example.py --python 3.12

# Add dependencies to the script
uv add --script example.py httpx

# Run the script
uv run --script example.py
```

ref:

<https://docs.astral.sh/uv/guides/scripts/#running-scripts>

example.py

```
# /// script
# dependencies = [
#     "requests<3",
#     "rich",
# ]
# ///

import requests
from rich.pretty import pprint

resp = requests.get("https://peps.python.org/api/peps.json")
data = resp.json()
pprint([(k, v["title"]) for k, v in data.items()][:10)])
```

Using tools

```
uvx ruff check
```

```
# with specific version
```

```
uvx ruff@0.3.0 check
```

```
# from github repo
```

```
uvx --from git+https://github.com/httpie/cli httpie
```

Install and run a tool

```
uv tool install ruff  
ruff check
```

Working on projects

Create a new project:

```
uv init hello-world  
cd hello-world
```

Initialize a project in the current directory:

```
mkdir hello-world  
cd hello-world  
uv init
```

ref: <https://docs.astral.sh/uv/guides/projects>

Creating a project for a library

```
uv init --lib
```

```
.  
├── README.md  
├── pyproject.toml  
└── src  
    ├── hello_world  
    │   ├── __init__.py  
    │   └── py.typed
```

```
3 directories, 4 files
```


pyproject.toml (PEP 621)

```
[project]
name = "hello-world"
version = "0.1.0"
description = "Add your description here"
readme = "README.md"
authors = [
    { name = "narumi", email = "toucans-cutouts0f@icloud.com" }
]
requires-python = ">=3.14"
dependencies = []

[build-system]
requires = ["uv_build>=0.9.30,<0.10.0"]
build-backend = "uv_build"
```

Add packages

```
# Add a package to the current project  
uv add httpx
```

```
# Add dev dependencies  
uv add --dev ruff ty pytest pytest-cov
```

```
# Add a git dependency  
uv add git+https://github.com/psf/requests
```

ref: <https://docs.astral.sh/uv/guides/projects/#managing-dependencies>

Remove packages

```
# Remove a package from the current project  
uv remove httpx
```

```
# Remove dev dependencies  
uv remove --dev ruff ty
```

References

- [Astral](#)
- [Astral blog](#)
- [uv docs](#)